

# Ethanol Blending in Gasoline – Finland

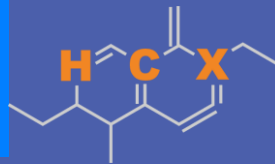
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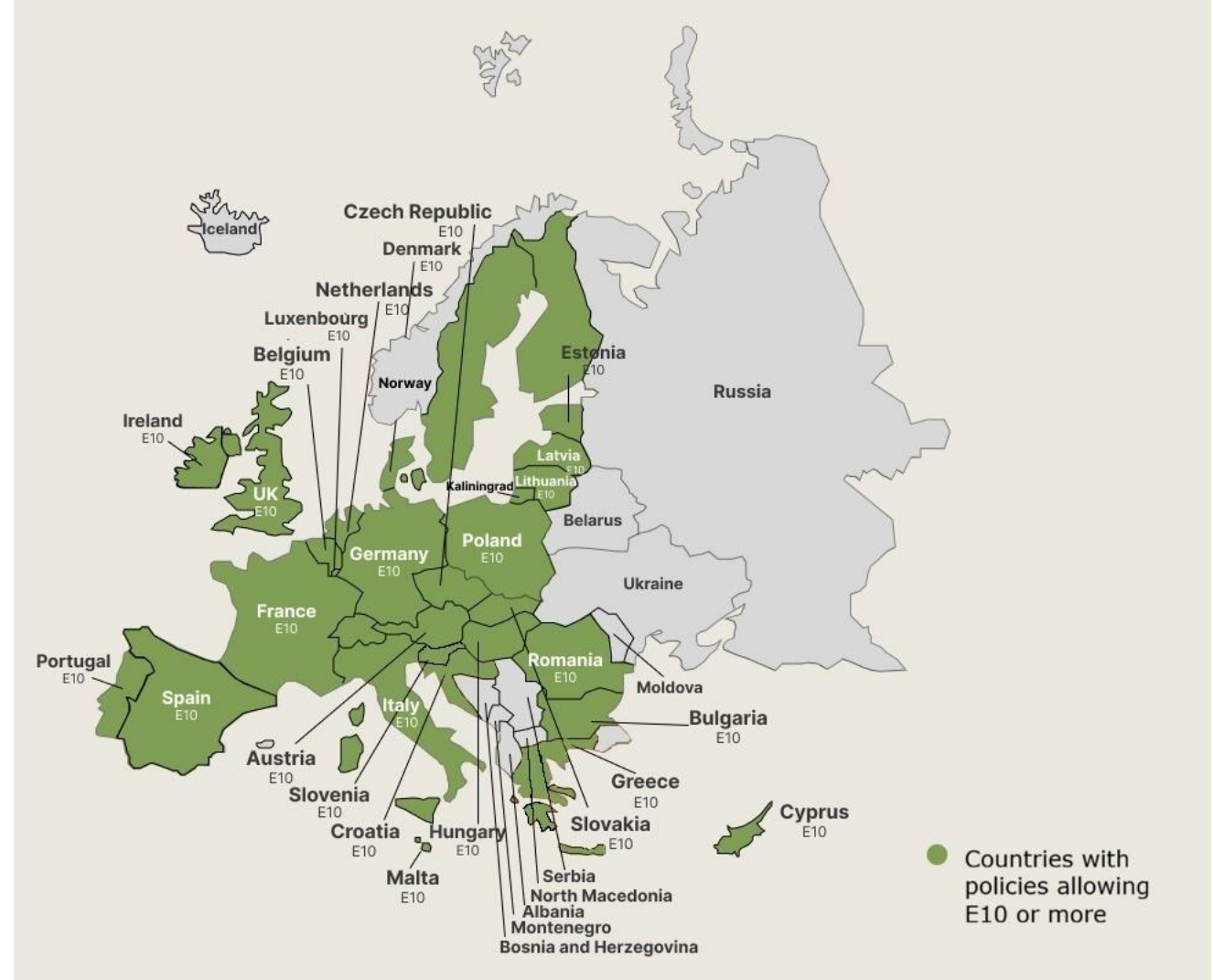


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# Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

# Ethanol Gasoline Blending in Finland

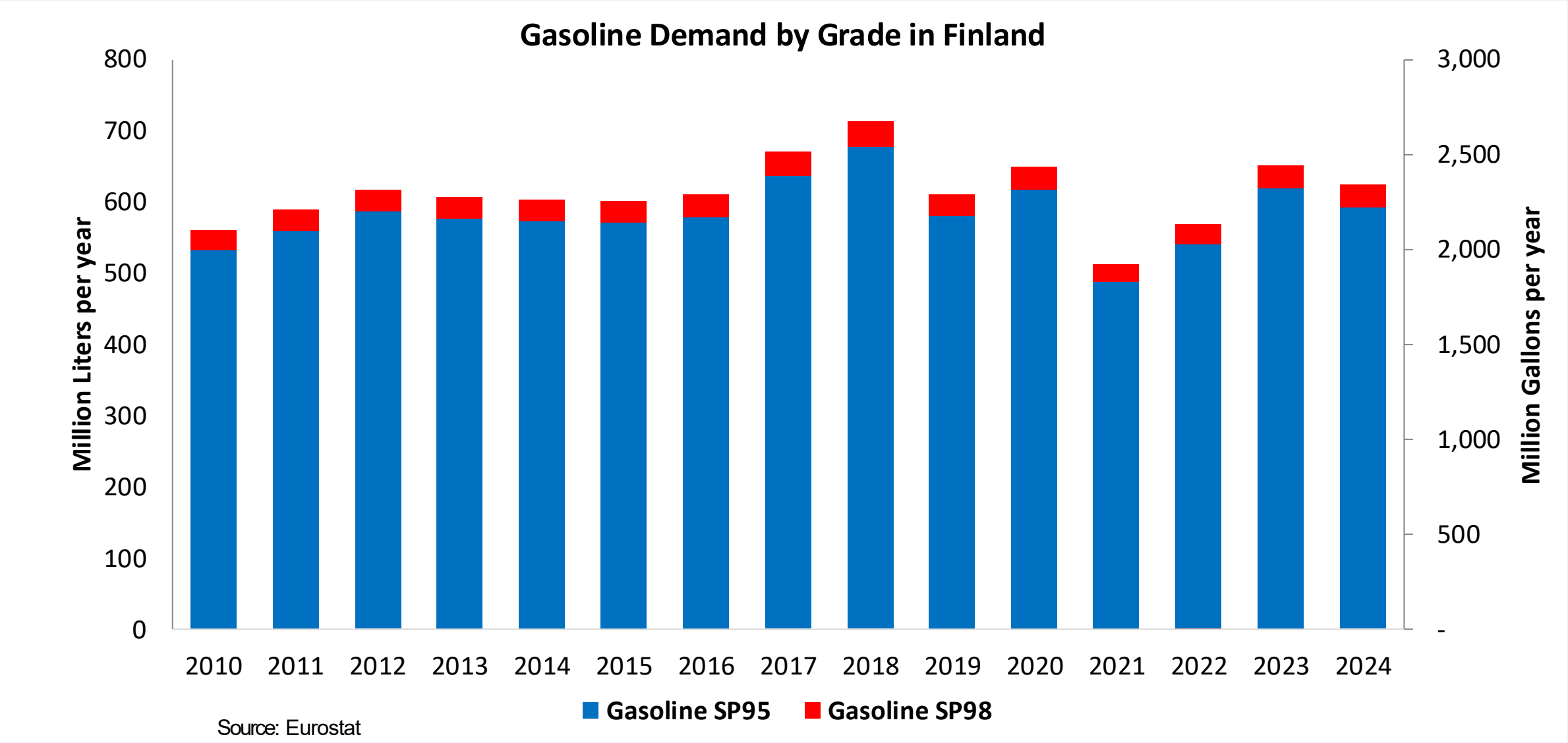


In 2024, gasoline consumption in Finland was 2,341 million liters (618 million gallons) and growth rate was 0.5%. Gasoline production and net imports were 5,450 and -3,109 million liters (1,440 and -821 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

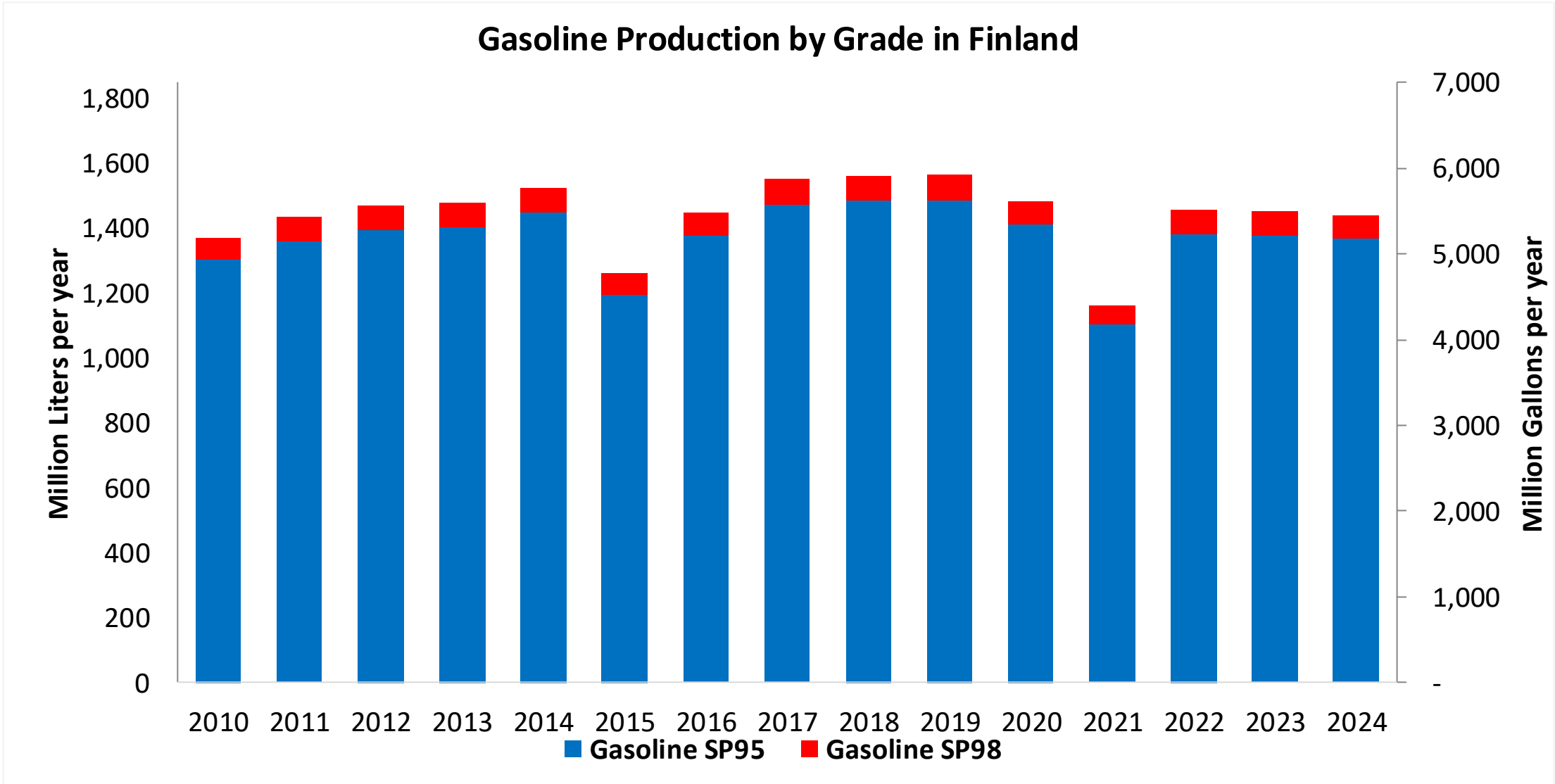
In 2024, ethanol consumption in Finland was 136 million liters (36 million gallons) representing 5.8% of gasoline demand. Ethanol production and Net Imports were 67 and 69 million liters (18 and 18 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

Source: Eurostat, European Commission

# Gasoline Demand in Finland

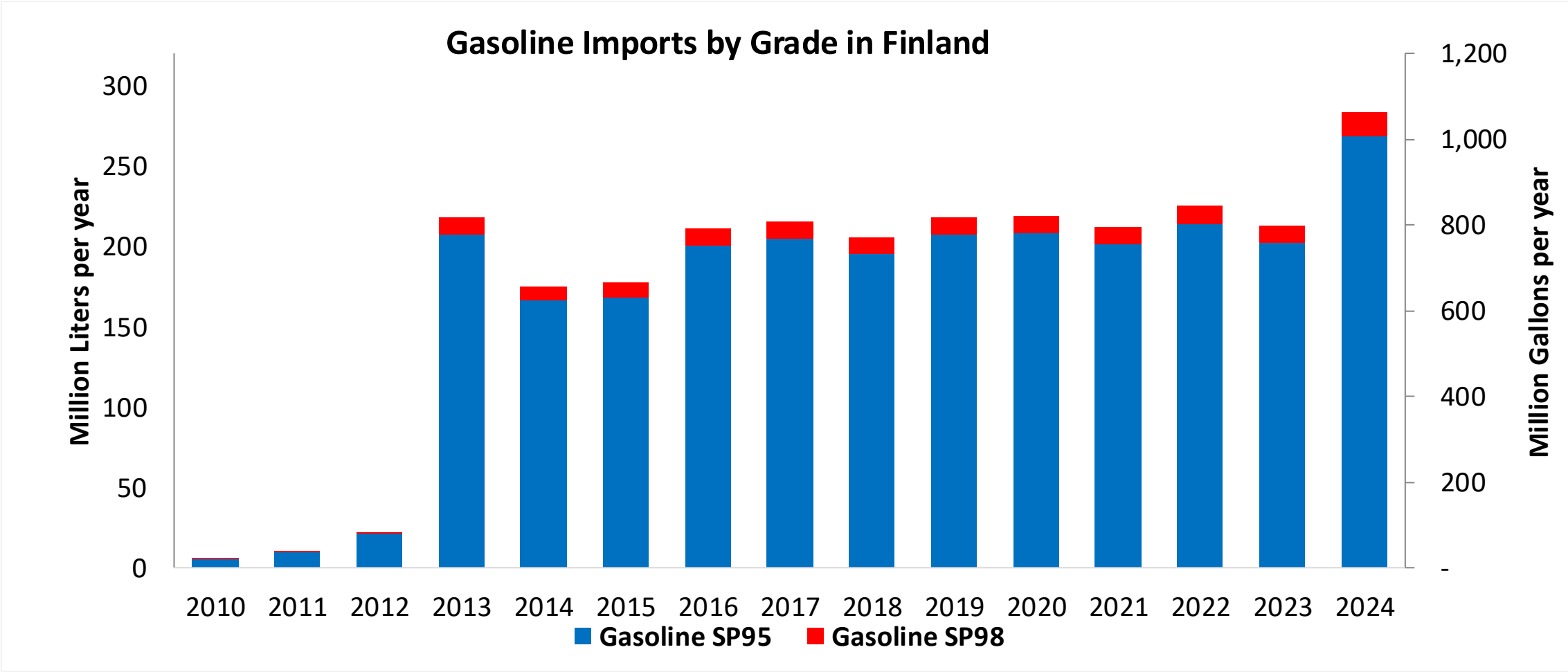


# Gasoline Production in Finland



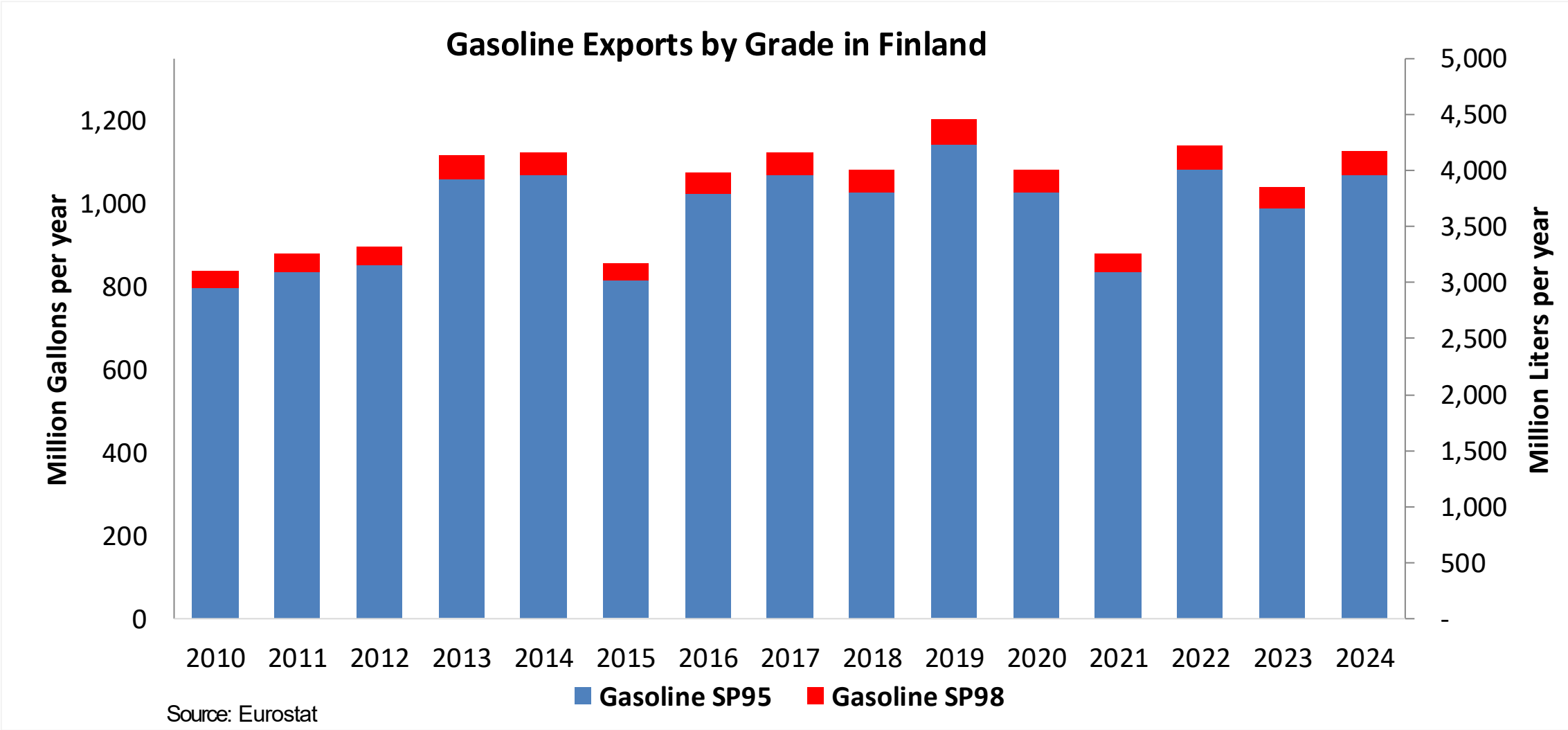
Source: Eurostat

# Gasoline Imports in Finland

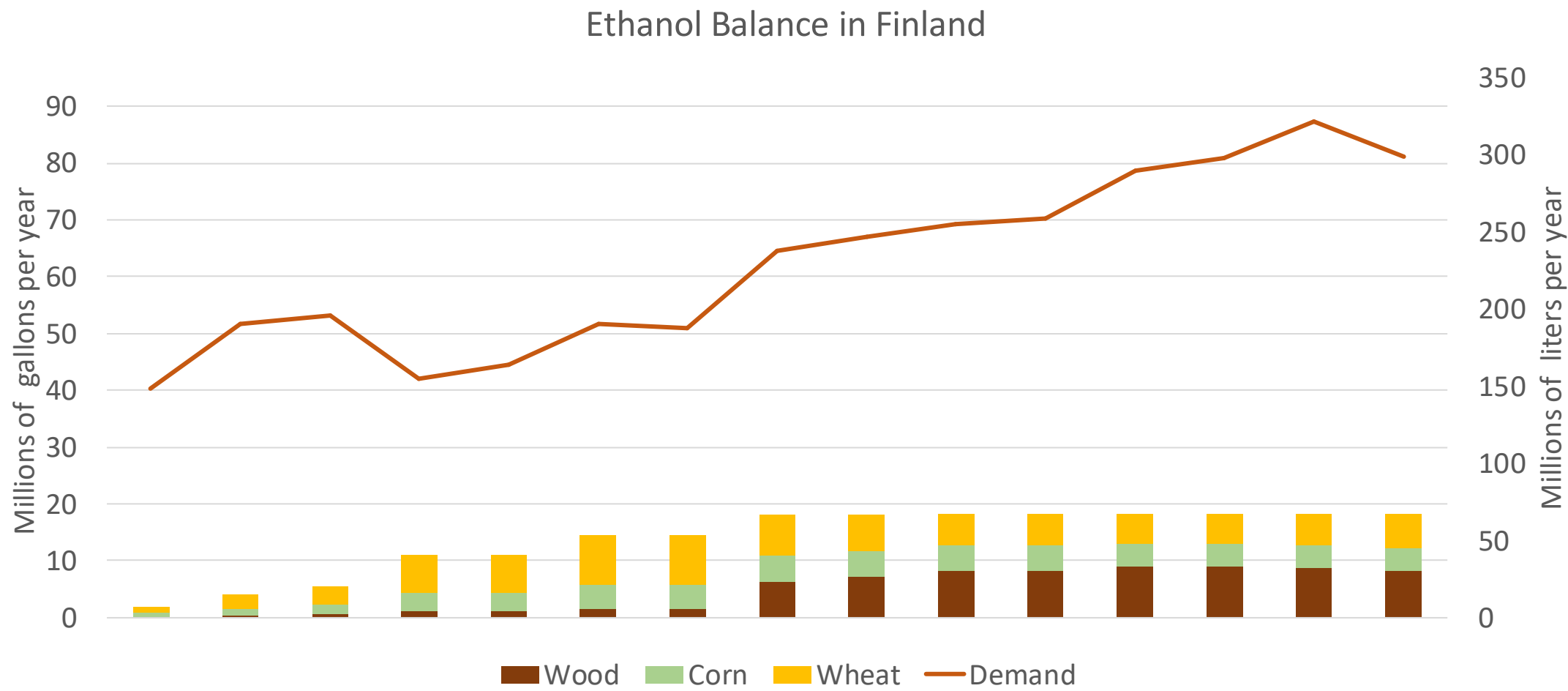


Source: Eurostat

# Gasoline Exports in Finland



# Ethanol Balance in Finland



Source: Eurostat

# Gasoline Quality in Finland

| Name                                | EN 228:2025 (Euro 6); Country Level Fuel Ordinances |       |            |       |           |       |            |       |
|-------------------------------------|-----------------------------------------------------|-------|------------|-------|-----------|-------|------------|-------|
| Implementation Date                 | 2025                                                |       |            |       |           |       |            |       |
| Applicability                       | All EU Contries + UK                                |       |            |       |           |       |            |       |
| Grade                               | RON 95 E5                                           |       | RON 95 E10 |       | RON 98 E5 |       | RON 98 E10 |       |
|                                     | Min                                                 | Max   | Min        | Max   | Min       | Max   | Min        | Max   |
| RON                                 | 95.0                                                |       | 95.0       |       | 98.0      |       | 98.0       |       |
| MON                                 | > 85                                                | > 88  | > 85       | > 88  | > 85      | > 88  | > 85       | > 88  |
| AKI                                 |                                                     |       |            |       |           |       |            |       |
| Benzene (%vol)                      | < 1                                                 |       |            |       |           |       |            |       |
| Aromatics (% vol)                   | < 35                                                |       |            |       |           |       |            |       |
| Olefins (%vol)                      | < 18                                                |       |            |       |           |       |            |       |
| Lead (g/L)                          | < 0.005                                             |       |            |       |           |       |            |       |
| Manganese (mg/L)                    | < 2,0                                               |       |            |       |           |       |            |       |
| Sulfur (ppm)                        | 10.0                                                |       |            |       |           |       |            |       |
| Oxygen (%w)*                        | < 2,7                                               | < 3,7 | < 2,7      | < 3,7 | < 2,7     | < 3,7 | < 2,7      | < 3,7 |
| Ethanol (%vol)                      | < 5                                                 | < 10  | < 5        | < 10  | < 5       | < 10  | < 5        | < 10  |
| PVR 37.8°C (psi)                    | <> 45 - 100 kPa in 6 Volatility Classes             |       |            |       |           |       |            |       |
| Overall Biofuels (%cal)             | 16.5                                                |       |            |       |           |       |            |       |
| Advanced Biofuels (%cal)            | 4.0                                                 |       |            |       |           |       |            |       |
| Bioethanol in Gasoline Sales (%cal) | -                                                   |       |            |       |           |       |            |       |
| GHG Emission Reduction (%)          | -                                                   |       |            |       |           |       |            |       |

\* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision



The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH<sub>2</sub>). The ring also has methyl groups at the 3 and 4 positions.

|                       |
|-----------------------|
| Catalytic Gasoline    |
| Light Primary Naphta  |
| Heavy Primary Naphta  |
| Reformate             |
| Isopentane            |
| Alkylate              |
| Isomerate             |
| Hydrocracked Gasoline |
| Coker Naphta          |
| n-Butane              |
| MTBE                  |
| ETBE                  |
| TAME                  |
| Aromatics             |

| Country     | Threat Category 1 (Dark Blue) | Threat Category 2 (Red) | Threat Category 3 (Grey) | Threat Category 4 (Purple) | Threat Category 5 (Light Grey) | Threat Category 6 (Orange) | Threat Category 7 (Yellow) | Threat Category 8 (Brown) | Threat Category 9 (Dark Blue) | Threat Category 10 (Light Yellow) | Threat Category 11 (Dark Purple) | Threat Category 12 (Dark Blue) |
|-------------|-------------------------------|-------------------------|--------------------------|----------------------------|--------------------------------|----------------------------|----------------------------|---------------------------|-------------------------------|-----------------------------------|----------------------------------|--------------------------------|
| Austria     | 15%                           | 15%                     | 15%                      | 15%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Belgium     | 18%                           | 18%                     | 18%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Bulgaria    | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Croatia     | 15%                           | 15%                     | 15%                      | 20%                        | 5%                             | 5%                         | 5%                         | 15%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Cyprus      | 15%                           | 15%                     | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Czechia     | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Denmark     | 15%                           | 15%                     | 15%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Estonia     | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Finland     | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| France      | 20%                           | 20%                     | 20%                      | 18%                        | 5%                             | 5%                         | 5%                         | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Germany     | 15%                           | 15%                     | 15%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Greece      | 15%                           | 15%                     | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Hungary     | 12%                           | 18%                     | 15%                      | 18%                        | 5%                             | 5%                         | 5%                         | 15%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Ireland     | 5%                            | 15%                     | 15%                      | 25%                        | 5%                             | 10%                        | 10%                        | 5%                        | 15%                           | 10%                               | 5%                               | 5%                             |
| Italy       | 15%                           | 15%                     | 15%                      | 20%                        | 5%                             | 5%                         | 5%                         | 15%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Latvia      | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Lithuania   | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Luxembourg  | 15%                           | 15%                     | 15%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Malta       | 15%                           | 15%                     | 15%                      | 20%                        | 5%                             | 5%                         | 5%                         | 15%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Netherlands | 18%                           | 18%                     | 18%                      | 18%                        | 5%                             | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Poland      | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Portugal    | 15%                           | 15%                     | 15%                      | 18%                        | 5%                             | 5%                         | 5%                         | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Romania     | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Slovakia    | 10%                           | 5%                      | 15%                      | 20%                        | 10%                            | 10%                        | 10%                        | 10%                       | 10%                           | 5%                                | 5%                               | 5%                             |
| Slovenia    | 15%                           | 15%                     | 15%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Spain       | 15%                           | 15%                     | 15%                      | 20%                        | 5%                             | 5%                         | 5%                         | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| Sweden      | 15%                           | 15%                     | 15%                      | 18%                        | 10%                            | 10%                        | 10%                        | 10%                       | 5%                            | 5%                                | 5%                               | 5%                             |
| UK          | 5%                            | 15%                     | 15%                      | 25%                        | 5%                             | 10%                        | 10%                        | 5%                        | 15%                           | 10%                               | 5%                               | 5%                             |

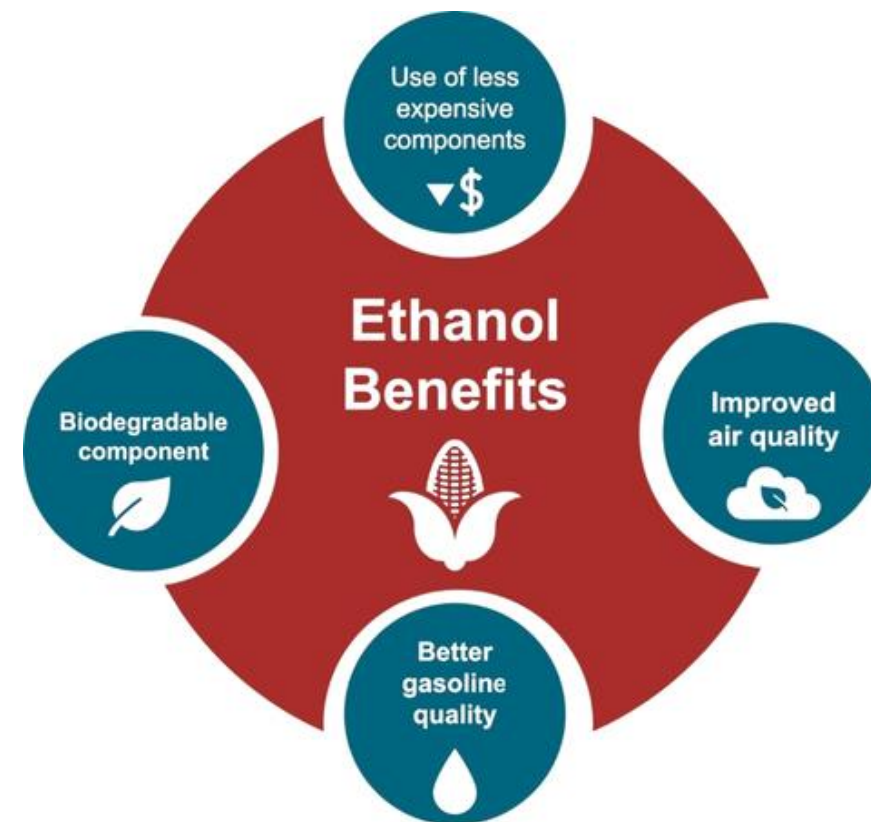
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

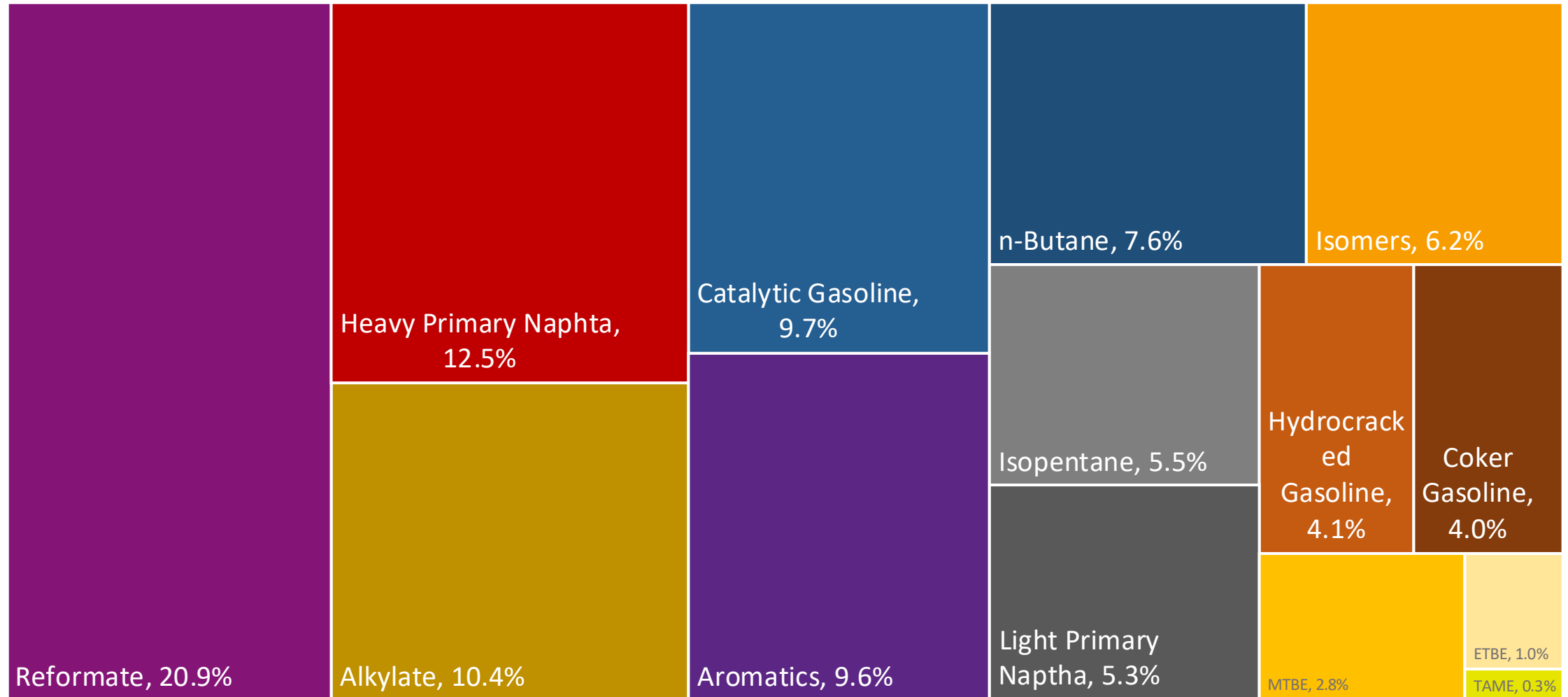
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

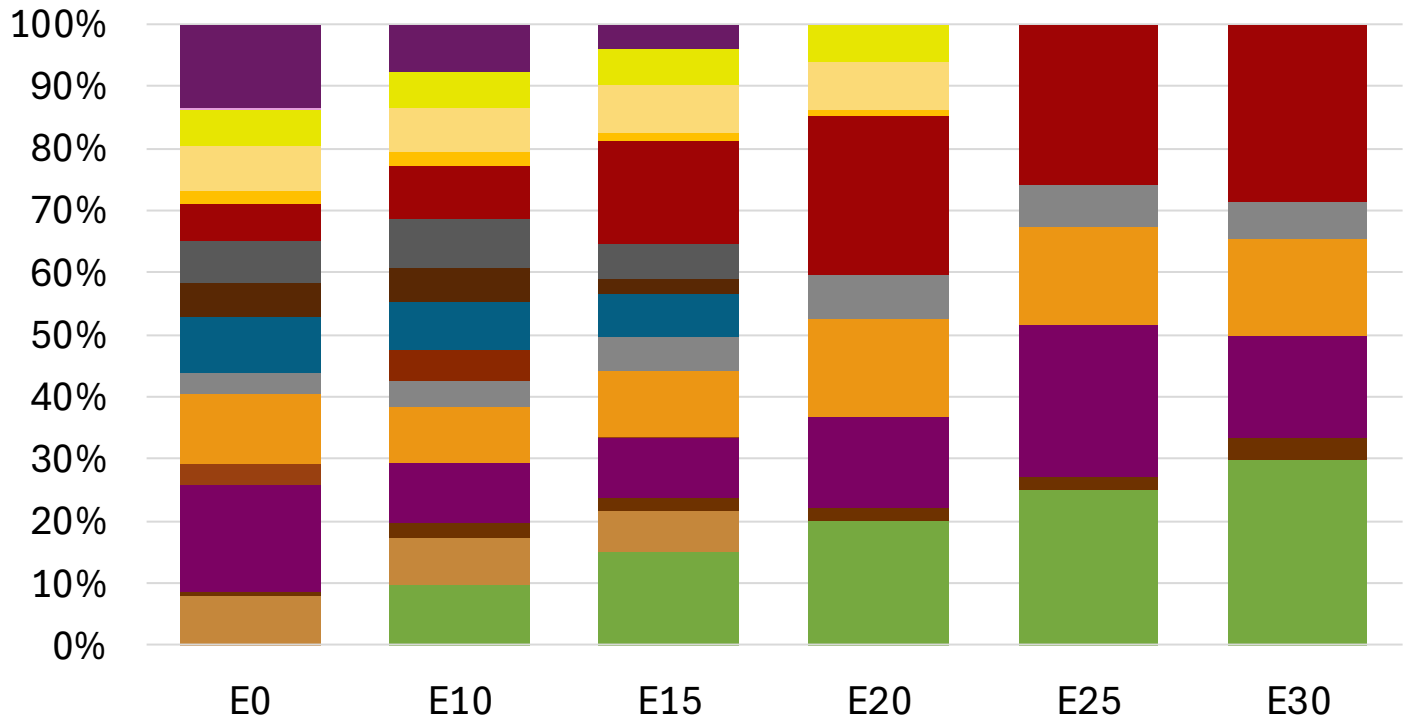
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



# Finland Available Blending Components



# Finland – Gasoline 95 – Constant Octane

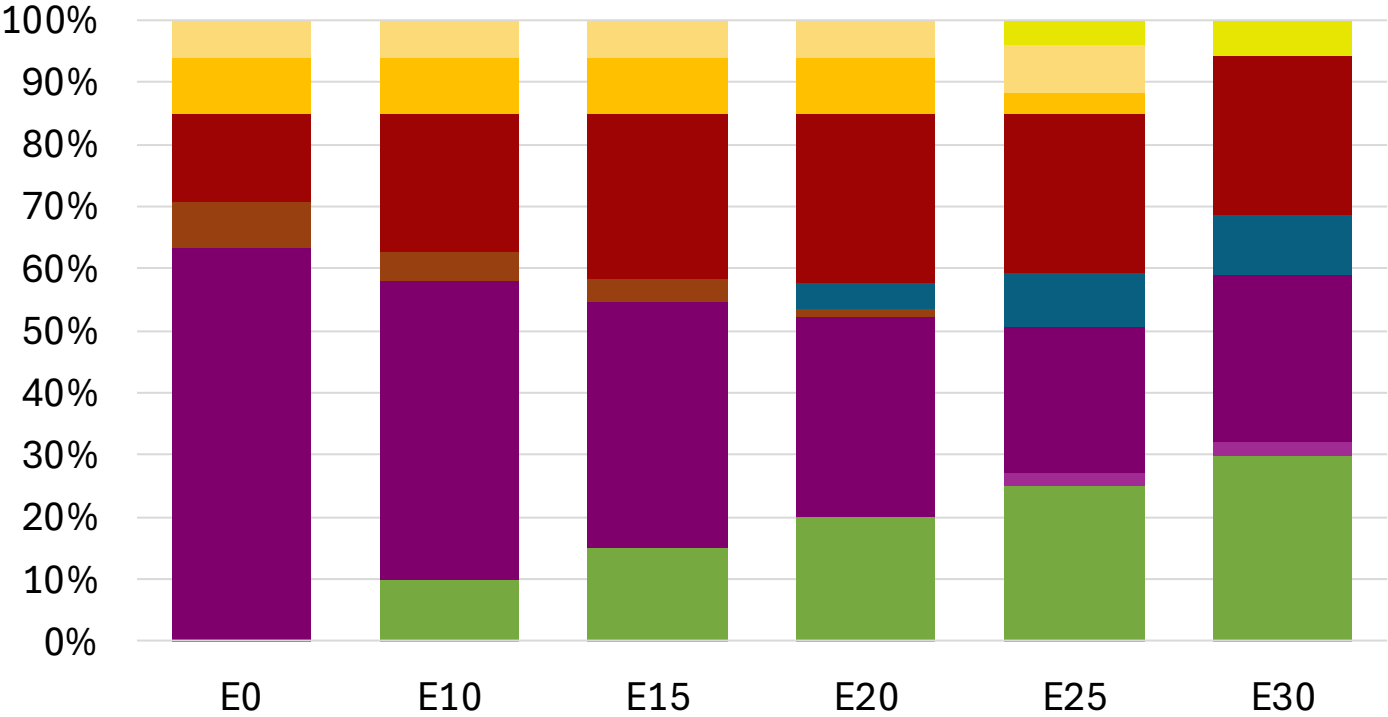


Average CIF 2024  
Prices

|                  |      |       |      |       |      |       |      |       |      |       |      |       |
|------------------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|
| Octane (RON)     | 95.0 |       | 95.0 |       | 95.0 |       | 95.0 |       | 95.0 |       | 95.8 |       |
| Price (US\$/gal) | \$   | 2.294 | \$   | 2.156 | \$   | 2.056 | \$   | 1.941 | \$   | 1.895 | \$   | 1.842 |

Source: Faro90

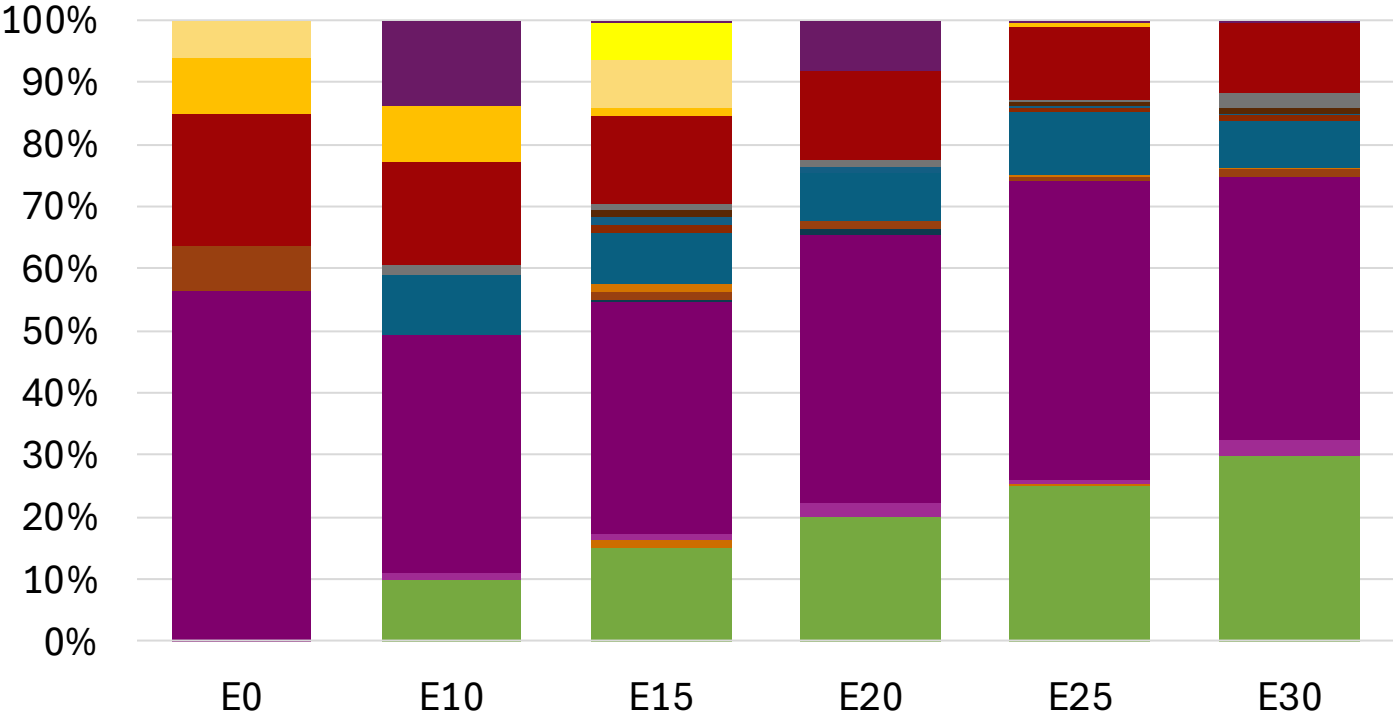
# Finland - Gasoline 98 - Constant Octane



Average CIF 2024  
Prices

|                     |          |          |          |          |          |          |
|---------------------|----------|----------|----------|----------|----------|----------|
| Octane<br>(RON)     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     |
| Price<br>(US\$/gal) | \$ 2.314 | \$ 2.197 | \$ 2.128 | \$ 2.056 | \$ 1.991 | \$ 1.941 |

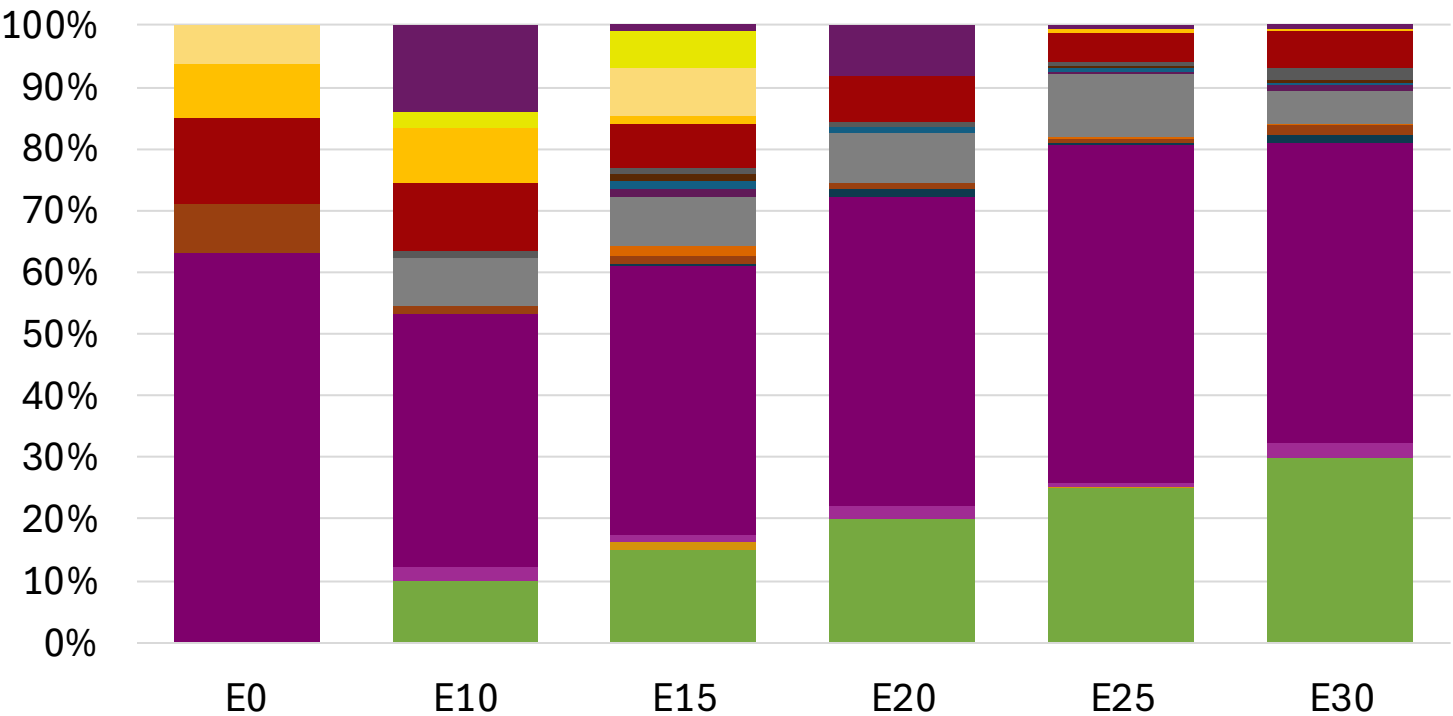
# Finland – Gasoline 95 – Increased Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 96.2     | 98.3     | 98.7     | 100.5    | 101.8    |
| Price (US\$/gal) | \$ 2.182 | \$ 2.182 | \$ 2.182 | \$ 2.182 | \$ 2.182 | \$ 2.182 |

# Finland – Gasoline 98 – Increased Octane

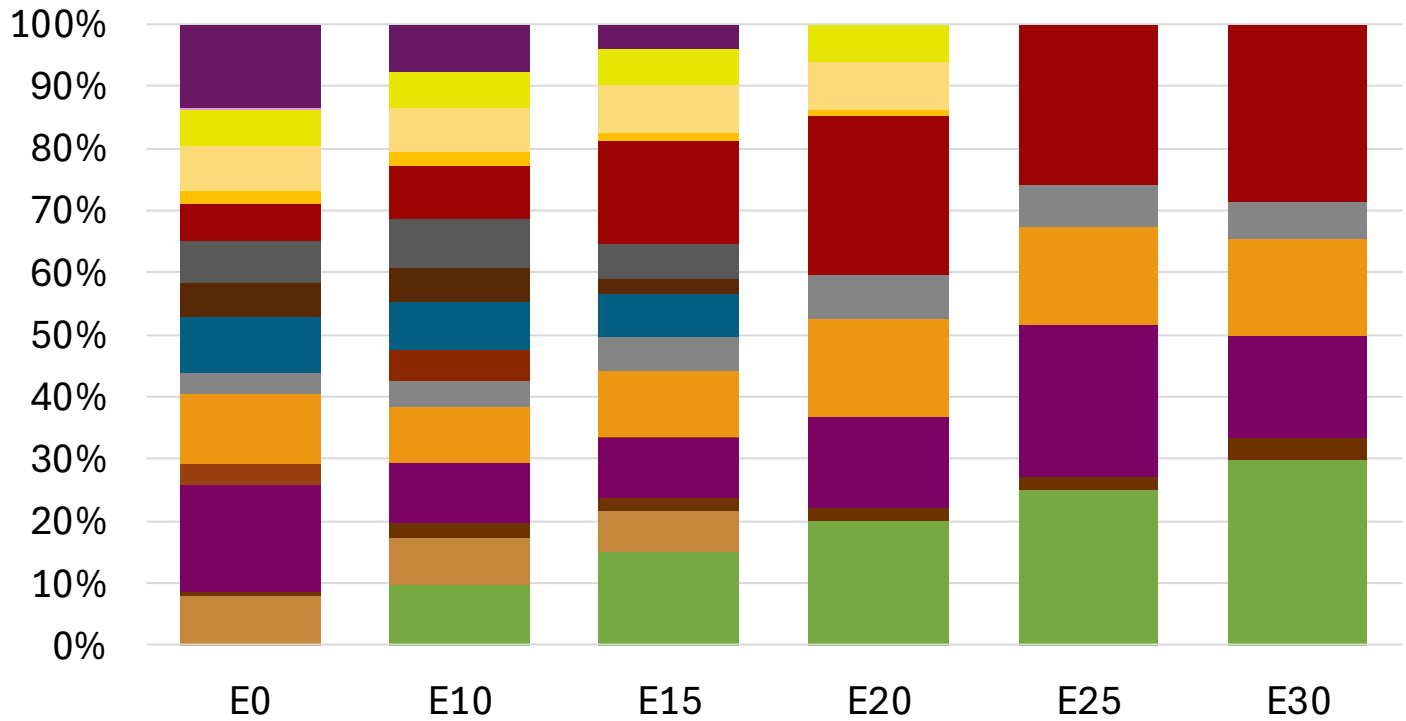


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 98.0     | 99.0     | 100.9    | 101.2    | 102.8    | 104.1    |
| Price (US\$/gal) | \$ 2.314 | \$ 2.314 | \$ 2.314 | \$ 2.314 | \$ 2.314 | \$ 2.314 |

Source: Faro90

# Finland – Gasoline 95 – Constant Octane

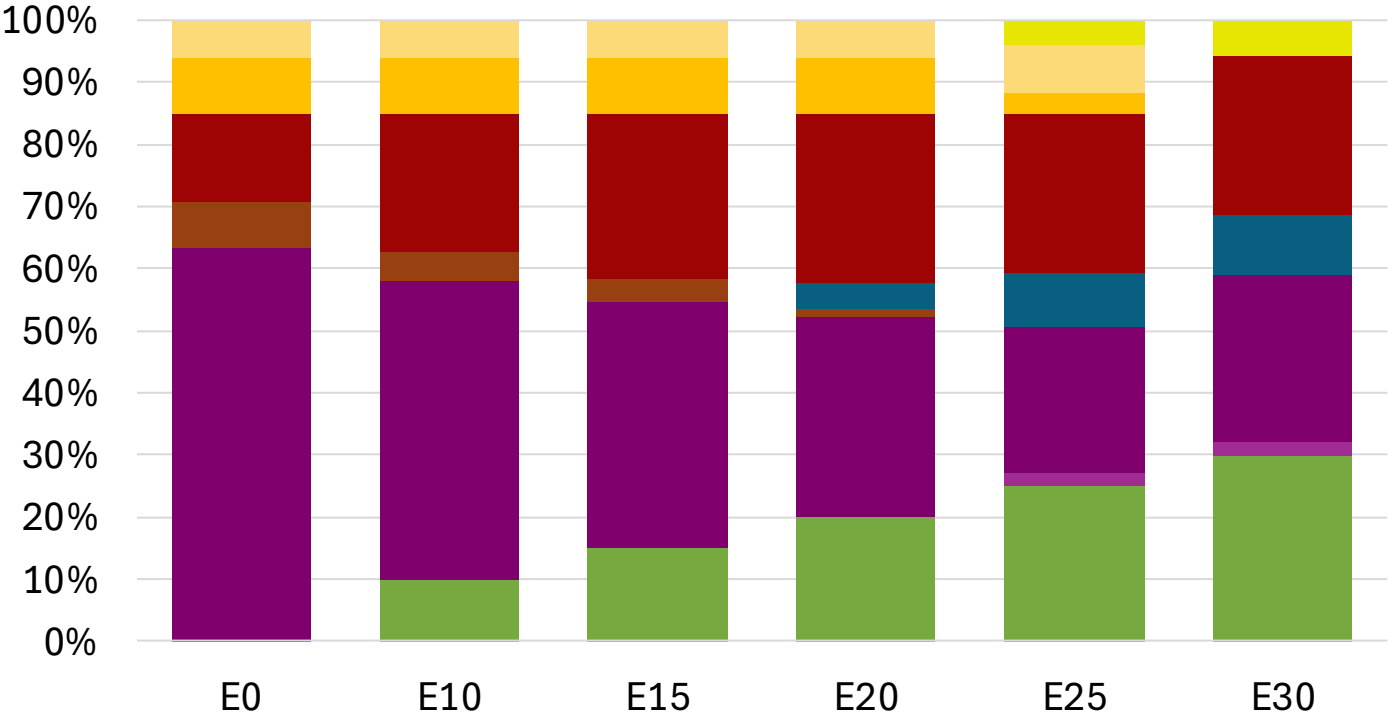
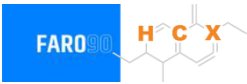


Average CIF 2024  
Prices

|                  |      |       |      |       |      |       |      |       |      |       |      |       |
|------------------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|
| Octane (RON)     | 95.0 |       | 95.0 |       | 95.0 |       | 95.0 |       | 95.0 |       | 95.8 |       |
| Price (US\$/gal) | \$   | 2.274 | \$   | 2.136 | \$   | 2.036 | \$   | 1.921 | \$   | 1.875 | \$   | 1.822 |

Source: Faro90

# Finland - Gasoline 98 - Constant Octane

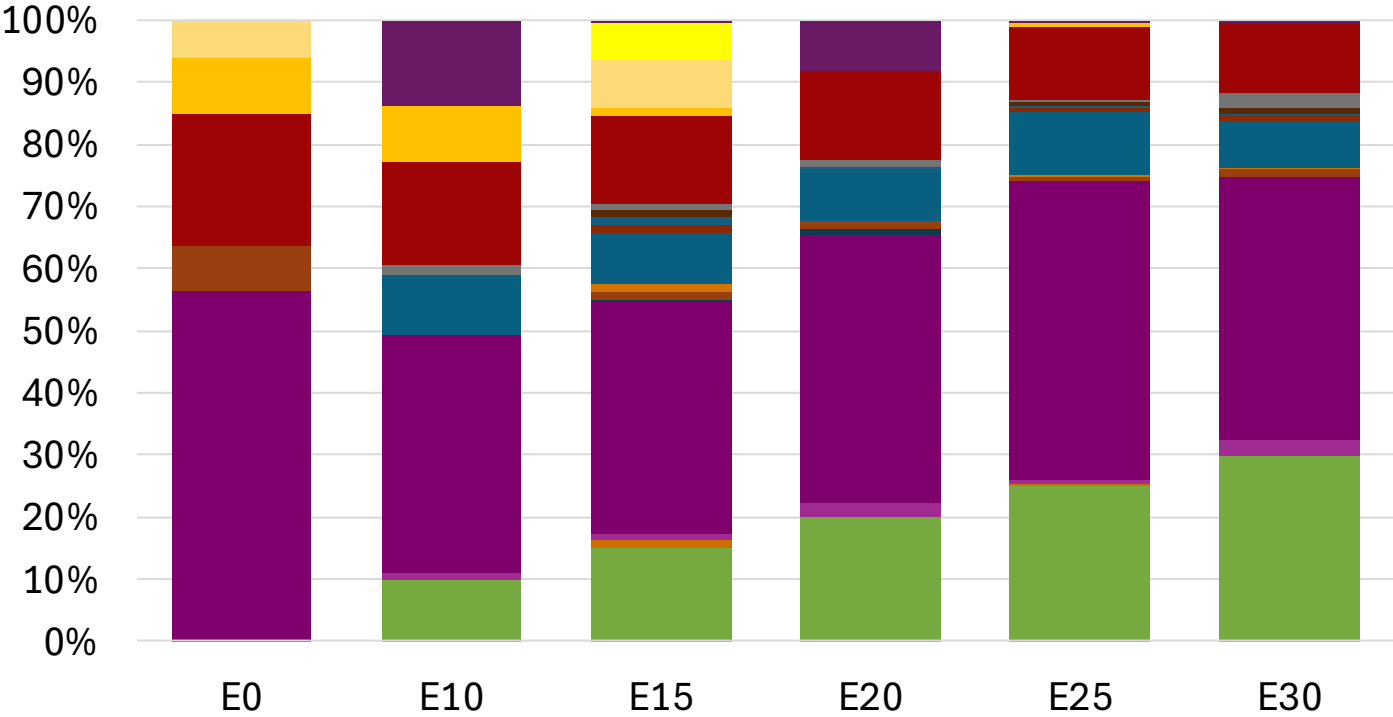
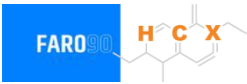


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     | 98.0     |
| Price (US\$/gal) | \$ 2.294 | \$ 2.177 | \$ 2.108 | \$ 2.036 | \$ 1.971 | \$ 1.921 |

Source: Faro90

# Finland – Gasoline 95 – Increased Octane

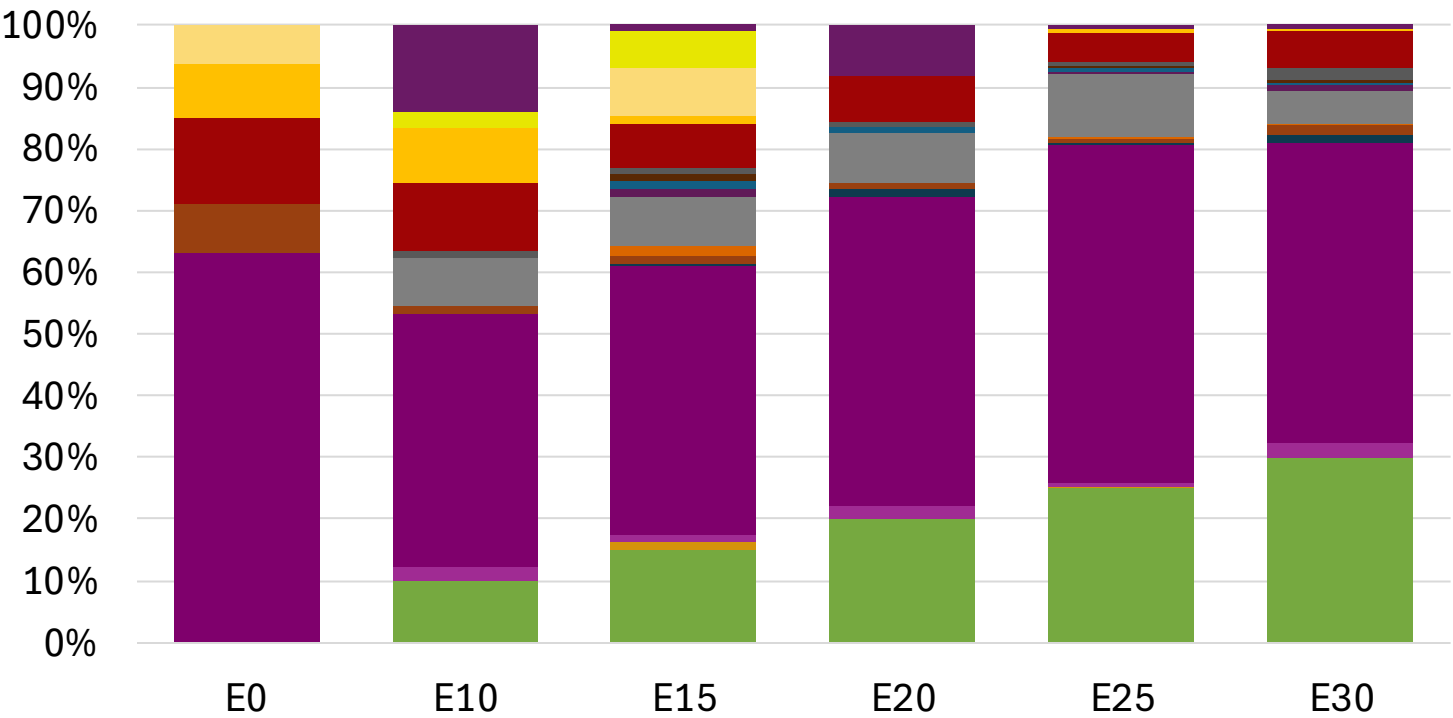
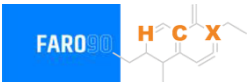


Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 95.0     | 96.2     | 98.3     | 98.7     | 100.5    | 101.8    |
| Price (US\$/gal) | \$ 2.162 | \$ 2.162 | \$ 2.162 | \$ 2.162 | \$ 2.162 | \$ 2.162 |

Source: Faro90

# Finland – Gasoline 98 – Increased Octane



Average CIF 2024  
Prices

|                  |          |          |          |          |          |          |
|------------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)     | 98.0     | 99.0     | 100.9    | 101.2    | 102.8    | 104.1    |
| Price (US\$/gal) | \$ 2.294 | \$ 2.294 | \$ 2.294 | \$ 2.294 | \$ 2.294 | \$ 2.294 |

Source: Faro90

# Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

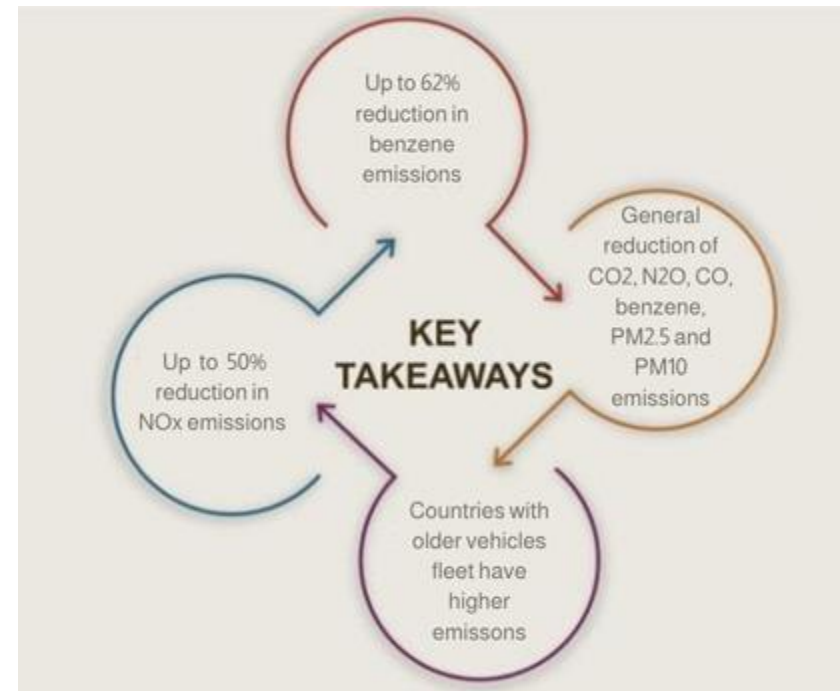
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

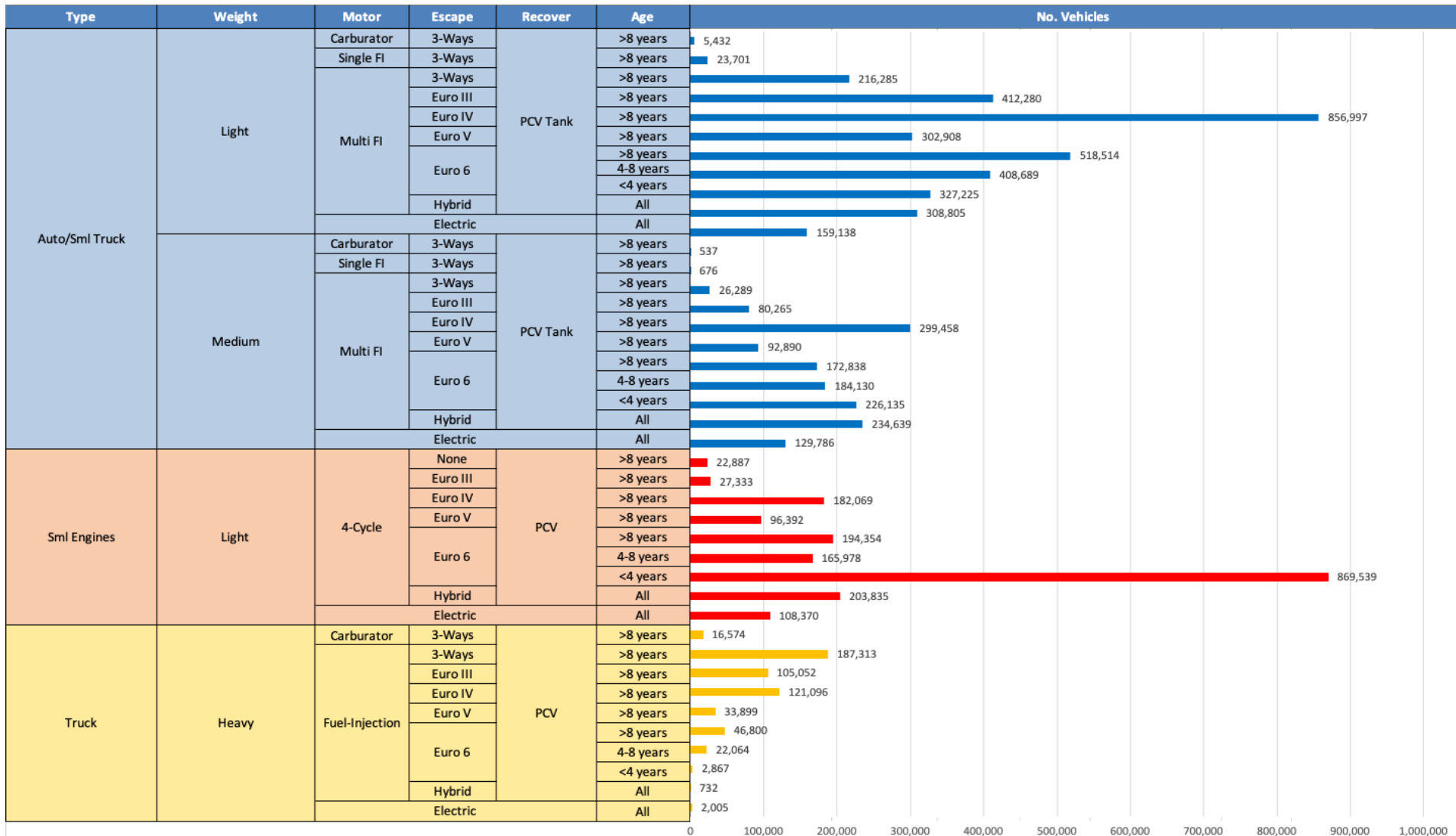
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet\*.

\*Source: Bureau of transportation statistics.

## Main Results



# Gasoline Vehicle Fleet - Finland

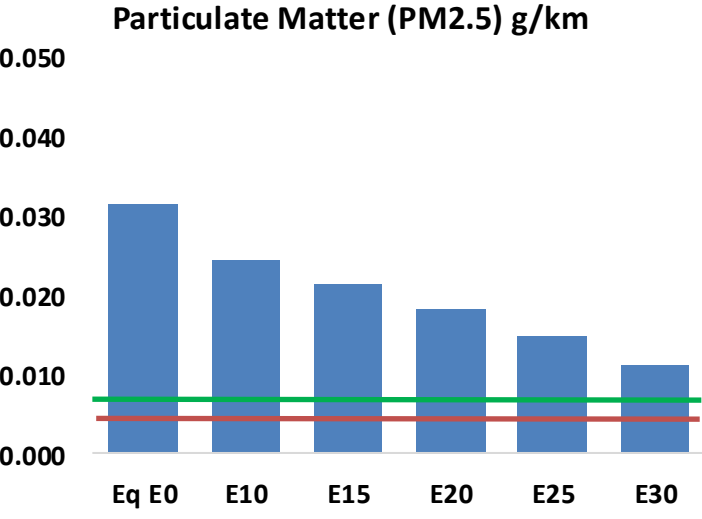
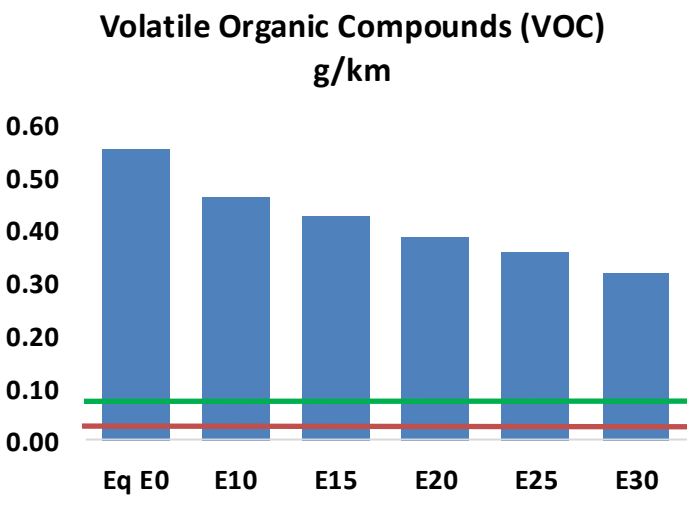
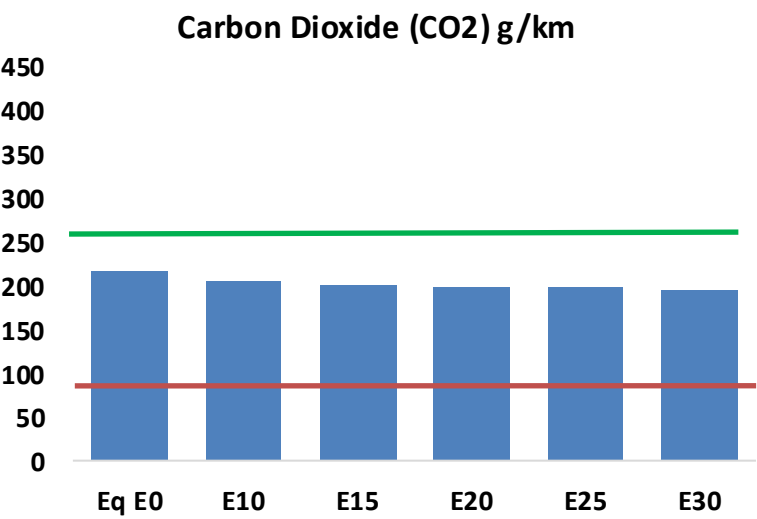
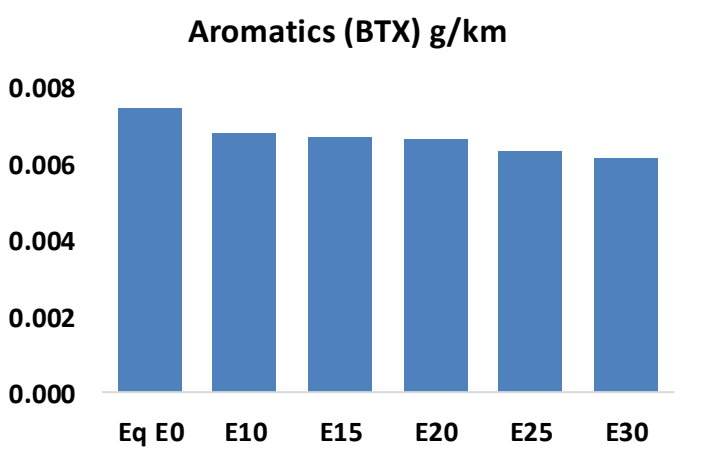
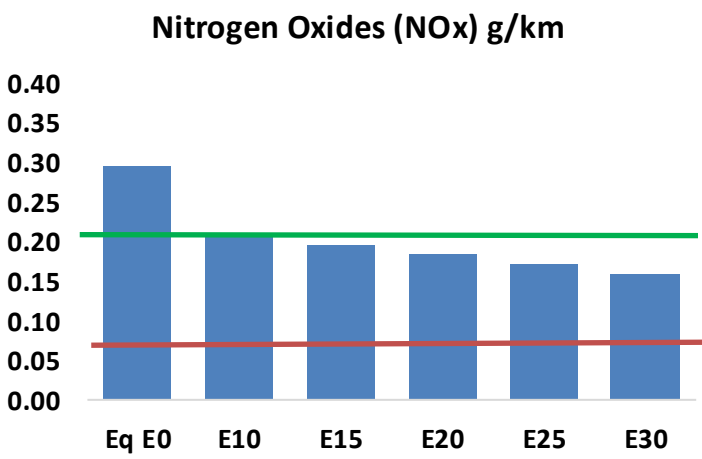
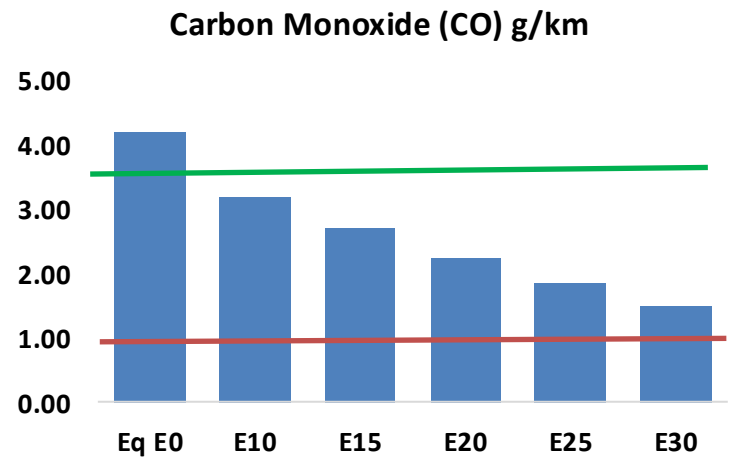
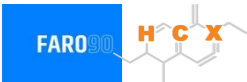


Vehicle Fleet: 7,396,775  
Gasoline Vehicle Fleet: 3,837,401

Source: Eurostat, ACEA

# France – Vehicular Emissions

TIER III BIN 125 United States  
Euro 6 Standard



# Finland – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0      | E5         | E10        | E15        | E20        | E25        | E30        | Reduction (%) |      |
|------------------------------------|------------|------------|------------|------------|------------|------------|------------|---------------|------|
|                                    |            |            |            |            |            |            |            | E10           | E20  |
| CO                                 | 469,869    | 408,719    | 347,570    | 292,577    | 234,989    | 190,028    | 145,320    | -26%          | -50% |
| CO2                                | 19,452,404 | 18,952,199 | 18,479,784 | 18,108,243 | 17,924,390 | 17,818,561 | 17,490,113 | -5%           | -8%  |
| VOC                                | 114,075    | 103,064    | 92,053     | 82,593     | 72,871     | 65,335     | 55,493     | -19%          | -36% |
| NOx                                | 27,517     | 20,638     | 19,262     | 18,154     | 17,121     | 15,982     | 14,777     | -30%          | -38% |
| SOx                                | 221        | 204        | 188        | 174        | 160        | 145        | 130        | -15%          | -28% |
| NH3                                | 7,866      | 7,788      | 7,710      | 7,747      | 7,834      | 7,895      | 7,946      | -2%           | 0%   |
| N2O                                | 324        | 322        | 321        | 323        | 329        | 333        | 337        | -1%           | 2%   |
| CH4                                | 22,977     | 22,977     | 22,977     | 23,322     | 23,828     | 24,264     | 24,678     | 0%            | 4%   |
| Lead                               | -          | -          | -          | -          | -          | -          | -          | 0%            | 0%   |
| Butadiene                          | 467        | 434        | 400        | 373        | 347        | 326        | 299        | -14%          | -26% |
| Acetaldehyde                       | 1,791      | 2,056      | 3,017      | 4,594      | 6,259      | 7,152      | 8,451      | 68%           | 249% |
| Formaldehyde                       | 7,151      | 6,952      | 8,104      | 9,516      | 9,969      | 11,029     | 12,006     | 13%           | 39%  |
| Benzene                            | 1,102      | 1,059      | 1,003      | 987        | 979        | 936        | 906        | -9%           | -11% |
| PM2.5                              | 1,230      | 1,092      | 954        | 828        | 703        | 571        | 433        | -22%          | -43% |
| PM10                               | 4,504      | 3,999      | 3,494      | 3,035      | 2,577      | 2,091      | 1,585      | -22%          | -43% |

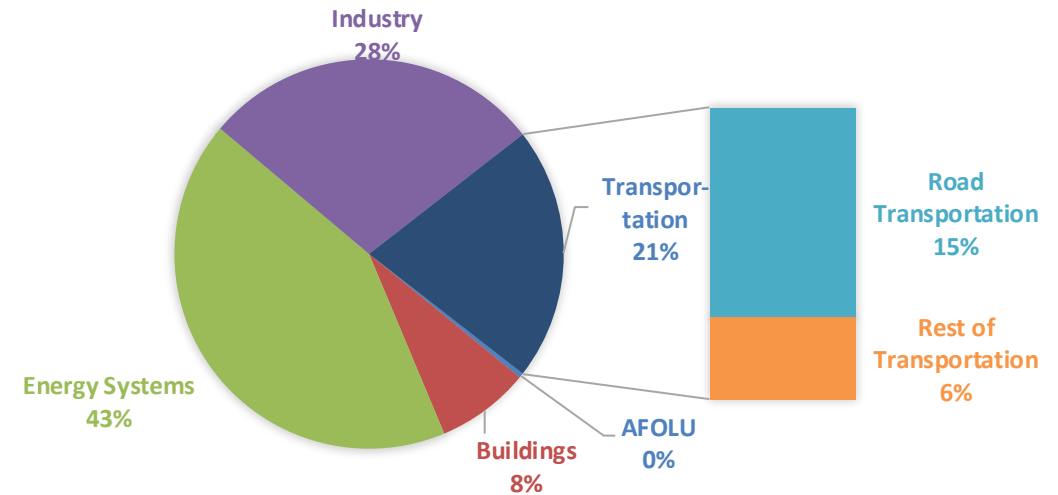
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

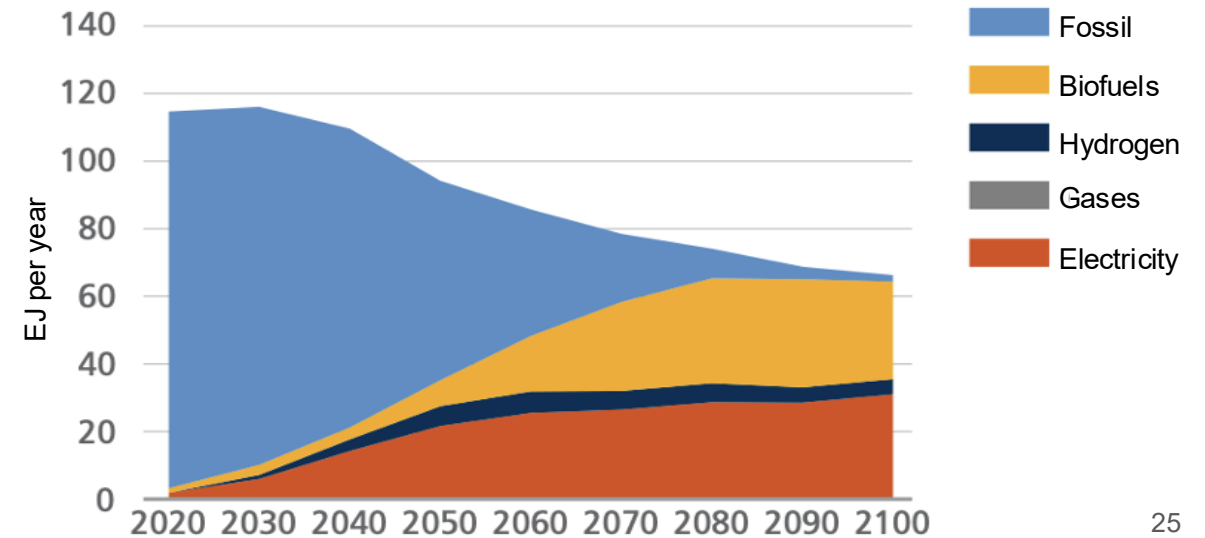
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

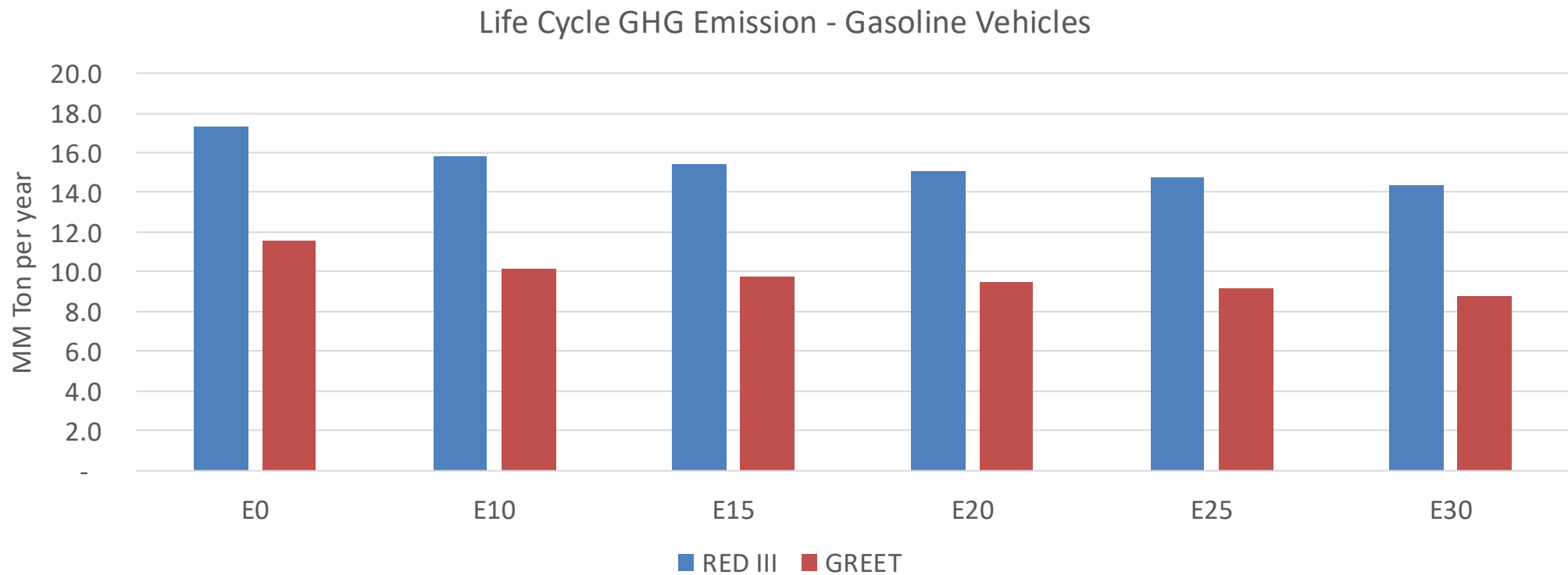
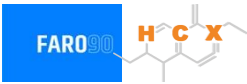
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



# Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



|                                  |      |
|----------------------------------|------|
| Country Emissions 2024 (MMT/año) | 64.9 |
| Target @ 2035 (MMT/year)         | 36.6 |
| Delta                            | 28.4 |

| %Reduction vs E0 | E0   | E10   | E15   | E20   | E25   | E30   |
|------------------|------|-------|-------|-------|-------|-------|
| GREET 2024       | 0.0% | 12.7% | 15.9% | 18.5% | 21.0% | 24.2% |
| RED II/III       | 0.0% | 8.9%  | 11.3% | 13.1% | 14.9% | 17.2% |

| % Target@2035 | E0   | E10  | E15  | E20  | E25  | E30   |
|---------------|------|------|------|------|------|-------|
| GREET 2024    | 0.0% | 5.2% | 6.5% | 7.6% | 8.6% | 9.9%  |
| RED II/III    | 0.0% | 5.5% | 6.9% | 8.0% | 9.1% | 10.5% |

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90